SET inertia

Inertia and the response of finite field capacity

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In SET, gravitational burden, translation and event progression share a finite field capacity:

$$S^2 + |\mathbf{V}|^2 + c^2\beta^2 = c^2.$$

Here V denotes motion in the specified spatial frame per reference time T, and $\beta = d\tau/dT$ records the probe's clock progression. Define the held-clock capacity by $\alpha^2 = 1 - S^2/c^2$. Then

$$\beta = \sqrt{\alpha^2 - \frac{|\mathbf{V}|^2}{c^2}}. \tag{1}$$

Consider a massive probe with constant, calibrated rest mass m, on the positive-clock branch $\beta > 0$. SET's retained probe coupling assigns the action

$$I_p = -mc^2 \int \beta \, dT, \qquad L_p = -mc^2\beta. \tag{2}$$

This connects the capacity relation to physical momentum. Differentiating with respect to velocity gives

$$\frac{\partial\beta}{\partial V_i} = -\frac{V_i}{c^2\beta}, \qquad P_i = \frac{\partial L_p}{\partial V_i} = \frac{mV_i}{\beta}. \tag{3}$$

The same expression that describes event progression therefore determines how much momentum accompanies a given motion. To examine inertia, hold the environmental burden S fixed and calculate the change in momentum produced by a change in velocity:

$$\mathcal{M}_{ij} \equiv \left(\frac{\partial P_i}{\partial V_j}\right)_S = \frac{m}{\beta}\left(\delta_{ij} + \frac{V_i V_j}{c^2\beta^2}\right). \tag{4}$$

This matrix is the probe's inertial response in these readings. A small velocity change satisfies $d\mathbf{P} = \mathscr{M}\, d\mathbf{V}$. Its transverse and longitudinal coefficients are

$$\mathcal{M}_\perp = \frac{m}{\beta}, \qquad \mathcal{M}_\parallel = \frac{m\alpha^2}{\beta^3}. \tag{5}$$

Both coefficients are positive. They distinguish changing the direction of motion from changing its speed. At fixed burden, as translation approaches the available capacity, $|\mathbf{V}| \to c\alpha$, the momentum required for further increases in speed diverges.

This gives inertia a specific capacity interpretation: **the response to a change in motion follows from the way translation and event progression share the same finite budget.** The invariant supplies that dependence; the retained action identifies its derivative with physical momentum.

The clock convention matters. A local held observer measures

$$\mathbf{u} = \frac{\mathbf{V}}{\alpha}, \qquad \gamma = \frac{\alpha}{\beta} = \frac{1}{\sqrt{1 - |\mathbf{u}|^2/c^2}}.$$

Consequently,

$$\mathbf{P} = m\gamma\mathbf{u}. \tag{6}$$

In local velocity units, the transverse and longitudinal coefficients become $m\gamma$ and $m\gamma^3$. At rest they both equal m. The reference-time coefficient m/α therefore describes a change of reading, rather than additional material mass.

The energy relation follows from the same action:

$$E = \mathbf{P} \cdot \mathbf{V} - L_p = \frac{mc^2\alpha^2}{\beta} = \alpha mc^2\gamma. \tag{7}$$

A held probe at the same burden has energy $E_{\text{held}} = mc^2\alpha$. Relative to that state, the kinetic increment is

$$E - E_{\text{held}} = \alpha mc^2(\gamma - 1) = \frac{m}{2\beta}\left[|\mathbf{V}|^2 + c^2(\beta - \alpha)^2\right]. \tag{8}$$

This is an exact nonnegative expression, equal to zero at rest. For small local speed, $|\mathbf{u}| \ll c$, it gives

$$E - E_{\text{held}} \simeq \frac{m|\mathbf{V}|^2}{2\alpha}, \qquad \frac{E - E_{\text{held}}}{\alpha} \simeq \frac{1}{2}m|\mathbf{u}|^2.$$

The familiar kinetic-energy relation is recovered in the observer's local units. Increasing speed requires increasing energy, while uniform motion in a homogeneous, stationary environment requires no continuing energy expenditure.

The same momentum expression also distinguishes acceleration from a changing environmental burden. In a fixed spatial frame,

$$\frac{d\mathbf{P}}{dT} = \mathcal{M}\frac{d\mathbf{V}}{dT} + \frac{mS\mathbf{V}}{c^2\beta^3}\frac{dS}{dT}. \tag{9}$$

The second term is essential when the probe crosses a burden gradient. A change in the surrounding capacity affects the momentum–velocity relationship even when the probe's rest mass remains fixed.

This connects inertia directly to the current reading used in the spherical construction. Along a specified sampling coordinate χ, define

$$J_\chi = -\frac{1}{2}\frac{dS^2}{d\chi}.$$

With m fixed and the burden varying along χ, differentiating equation (7) in momentum variables gives

$$dE = \mathbf{V}\cdot d\mathbf{P} + \frac{m}{\beta}J_\chi\, d\chi. \tag{10}$$

The coefficient multiplying the burden contribution is exactly the transverse inertial coefficient in equation (5). These two responses follow from one retained capacity coupling; they do not require separately adjusted coefficients.

The physical picture is one of conservative motion through a capacity-dependent environment. Translation changes the momentum and energy associated with the probe. Changes in gravitational burden alter those relationships through the same underlying expression. In the stationary free-motion problem, the conserved reference energy determines how these changes balance.

This establishes an account of **inertial response within SET**. The rest-mass scale remains calibrated, and the calculation does not yet identify inertia as a reaction against previously exported burden. That possible mechanism would have to reproduce the momentum and energy relations derived here through an explicit calculation of the evolving field.